BaiZe: From-Scratch Vision Encoder and Multimodal Grounding for Visual EDA Tasks

Abstract

Agentic assistants for electronic design automation (EDA) are already deployed in-house, but a correctness-critical slice of the flow—custom and analog design—resists them, because much of the correctness signal lives in *visual* artifacts: layout, schematic, and waveform views that a text-only model cannot observe.

We present the multimodal extension of **BaiZe**, a cost-efficient EDA agent built on the BaiZe language model [5]: a vision encoder pre-trained from scratch, aligned to the language backbone through a lightweight projector, and adapted to the semiconductor domain via reinforcement learning.

A larger ViT-L/14 encoder reaches 68.2% linear-probe on an internal protocol ($+3.9$ pp over ViT-B/16), yet the aligned MLLM averages only 26.1% over 11 multimodal benchmarks, indistinguishable from the smaller tower: encoder-quality gains do not transfer downstream at the 224-px single-tile setting.

We localise this gap and evaluate the full system on three tiers—12 general multimodal benchmarks, 4 domain (EDA) benchmarks, and the multimodal subset of an EDA flow benchmark—measuring whether a 196-token-per-image budget suffices for detail-dense views, and whether domain reinforcement learning closes the transfer gap.

We release the full recipe and an evaluation protocol, and report all measurements as **[TBD]**.

1 Introduction

Agentic assistants for electronic design automation (EDA) are already deployed in-house, coordinating design flows through Flash-tier masters. Yet a correctness-critical slice of the flow resists them: in custom and analog design, the signal that determines whether a layout "works" lives in *visual* artifacts—layout, schematic, and waveform views—that a text-only model cannot observe. A subagent that can only read scripts and logs is blind to the very object it is designing, and must rely on an engineer to translate geometry into text—a bottleneck that scales with design volume.

Contributions. We build the multimodal core of **BaiZe**—a cost-efficient EDA agent—on top of the BaiZe language backbone [5], and ask a single question: *can a from-scratch vision encoder make such a small model see the visual EDA artifacts that carry the correctness signal?*

- **A from-scratch vision encoder.** BaiZe pre-trains its own vision encoder rather than reusing an off-the-shelf one, and selects the tower from iso-parameter candidates at the 500–600 M scale. A ViT-L/14 tower reaches 68.2% linear-probe on an internal protocol, $+3.9$ pp above ViT-B/16, at a 196-token-per-image budget.
- **Cost-bounded multimodal alignment.** BaiZe connects the encoder to the frozen language backbone through a lightweight projector, following the staged curriculum of open multimodal assistants, and asks whether a 196-token-per-image budget suffices for detail-dense layout and schematic views. We show that encoder-quality gains do *not* transfer downstream:

the larger tower averages 26.1% over 11 benchmarks, indistinguishable from the smaller one, and we localise the gap to the 224-px single-tile setting and the alignment architecture.

- **Domain adaptation.** A final reinforcement-learning stage adapts the multimodal model to proprietary semiconductor practice, learning team conventions that no public benchmark measures.
- **Three-tier evaluation.** We evaluate end-to-end on 12 general multimodal benchmarks, 4 domain (EDA) benchmarks, and the multimodal subset of an EDA flow benchmark, under a single frozen execution-based protocol.

2 Background and Related Work

2.1 Multimodal Vision-Language Models

Vision-language models connect a vision encoder to a language model through a lightweight projector, enabling end-to-end visual understanding and instruction following. LLaVA [3] established visual instruction tuning—aligning a vision encoder to a frozen LLM via a linear projection followed by instruction fine-tuning—as a scalable recipe for open multimodal assistants. LLaVA-OneVision [2] consolidates this line into a single open model spanning single-image, multi-image, and video tasks through a staged curriculum: stage-1 projector pre-training, stage-1.5 mid-training, and stage-2 instruction tuning. BaiZe follows this curriculum in stage (iv) (Section 5), substituting a from-scratch vision encoder (stage iii) for the frozen off-the-shelf encoder.

2.2 LLMs for Electronic Design Automation

A growing line of work applies LLM agents to chip design tasks such as register-transfer-level generation, verification, and bug triage [4, 1]. These applications motivate a cheap, always-on, agentic multimodal core that can run per-team or per-engineer; our work supplies the efficient multimodal backbone such systems require.

3 Architecture and Pipeline Overview

BaiZe is a multimodal EDA agent assembled in three stages on top of the BaiZe language backbone [5]: (i) from-scratch pre-training of a custom vision encoder, (ii) alignment of the encoder and the backbone into a multimodal large language model, and (iii) semiconductor-domain adaptation via reinforcement learning.

This section fixes the shared scaffolding—the vision encoder, the alignment projector, and the global training environment—before each stage is detailed in its own section.

The language backbone is the BaiZe $\approx$2.2B hybrid model, fixed and imported from its companion paper [5]; we do not re-train or re-describe it here, and keep it frozen throughout every stage.

3.1 Vision Encoder

The vision encoder is pre-trained from scratch in stage (i) and is selected from iso-parameter candidates at the 500–600 M scale; Section 4 reports the comparison and its outcome.

3.2 Pipeline and Global Environment

Table 1 summarises the three-stage pipeline. The vision encoder and alignment stages run on a single $8\times$H100 node (CUDA 12.8, PyTorch 2.8.0, Python 3.10); the aligned multimodal model uses a 196-token-per-image budget throughout.

4 Vision Encoder Pretraining

Stage (i) pre-trains a custom vision encoder from scratch; we summarise the outcome here and defer the design study to a companion report.

Table 1: BaiZe multimodal pipeline overview.

Stage	Task	Status	Section
(i)	Vision encoder	complete	4
(ii)	MLLM alignment	in progress	5
(iii)	Domain adapt. (RL)	design	6

Selected encoder. We adopt **OpenVision2**, a pure-attention ViT at width-512/depth-30 (126.8M parameters), trained from scratch with a bidirectional trunk at 224px/16-px patches on a $\approx$18.5M-pair image–text corpus (CC12M + Amshaker). The objective combines a global InfoNCE contrastive term with a caption-*independent* masked-patch reconstruction term at a 1:1 weight and a 0.6 mask ratio.

Final result. On the frozen trunk with an ImageNet-1k linear probe, the encoder reaches $11.50\% \rightarrow \mathbf{19.76}\%$ top-1 from 5.1M to 55.3M samples (CC12M+Amshaker, 18.5M unique pairs), and 20.27% at 176M samples on a 58.8M-pair corpus (adding GPIC), with no saturation. This is $+12.4$ pp above a contrastive-only baseline at the same sample budget, and it is the operating point used by stage (ii). The $224/16$ resolution is retained because it is best on both loss and throughput; whether 196 visual tokens suffice for detail-dense layout and schematic views is tested in stage (ii) (Section 5).

Encoder upgrade. A second tower—ViT-L/14 (304M) trained with open_clip under the same objective but lower peak LR (5×10^{-4}) and longer warm-up (3000 steps)—reaches $\mathbf{68.236 \pm 0.018}\%$ top-1 (5-NN 58.904%) on ImageNet-1k (Protocol B, 3 seeds), versus 64.337% / 55.956% for the ViT-B/16 (86M) tower; the $+3.9$ pp gap is a joint effect of tower size, LR, and warm-up, not a capacity-only attribution.

5 MLLM Alignment

Stage (ii) connects the vision encoder (Section 4) to the language backbone via a lightweight projector, following the staged curriculum of open multimodal assistants; layout and schematic views account for 70.2% of benchmark tasks [6], so a multimodal core shares the same execution-based protocol as the text-only stage [5].

Tasks and curriculum. Three task families each admit objective ground truth: (i) *image→script* (rendered view → PyAether/SKILL script, self-labelled by execution); (ii) *script→image* (predict geometry, self-labelled); (iii) *layout VQA* (attribute questions, expert-labelled). The first family shares the assertion sandbox of the text-only stage [5], giving a common metric. We run stage-1 projector pre-training on captions, stage-1.5 mid-training on the three families, and stage-2 instruction tuning on the full mixture; ablations vary the projector type, tower freezing, and visual-token count. The $224/16$ setting (196 tokens/image) is retained; whether it transfers to detail-dense line drawings is what stage (ii) tests.

Downstream results. Table 2 reports the Stage-2 step-3500 checkpoint on 12 lmms-eval benchmarks for ViT-L/14 (304M) and ViT-B/16 (86M). The 11 accuracy benchmarks average to 26.10% vs. 26.12% (-0.02 pp, indistinguishable); despite the $+3.9$ pp encoder advantage (Section 4), downstream metrics are flat, suggesting the bottleneck lies not in tower capacity but in the 224-px single-tile setting, the MLLM architecture, or weak proxy-to-downstream transfer.

6 Semiconductor Domain Adaptation

Stage (iii) completes the pipeline by adapting the multimodal model to proprietary design practice. It differs from the verifiable-reward stage of the text-only recipe [5] in two respects, which we state explicitly because otherwise the two stages would coincide: the reward is no longer verifiable but *learned from human judgement*, and the target behaviour is no longer general API usage but a team's internal conventions—naming rules, layer policies, review checklists—which no public benchmark measures.

Table 2: MLLM downstream results (Stage-2 step 3500, %; internal lmms-eval, single seed, 224-px single-tile). MME† is excluded from the 11-set mean.

Benchmark	ViT-L	ViT-B	Benchmark	ViT-L	ViT-B
AI2D	60.65	60.49	CountBench	10.59	9.98
RealWorldQA	45.36	46.41	DocVQA	7.35	7.44
MMMU_val	44.00	42.67	OCRBench	1.60	1.60
BLINK	38.88	39.59	InfoVQA	16.18	16.59
MMStar	27.17	27.07	ChartQA	12.16	12.92
MMMU_Pro	23.12	22.60	**11-set avg**	**26.10**	**26.12**
MME† (perc./cog./total): 513.75/204.29/718.04				559.67/246.79/806.45	

Method. We collect pairwise judgements from in-house design reviews over the stage-(ii) task families, fit a domain reward model on them, and optimise the aligned model against that model under a KL-regularised objective. Because preference data is far scarcer than the verifiable signal of the text-only stage [5], we initialise from the stage-(ii) checkpoint and constrain the update rather than training the objective from the pre-trained model.

Why a small model. The rationale for a $\approx$2.2B core is strongest here, and it is a constraint rather than a preference. Proprietary layouts and proprietary preference data cannot leave the premises, so the choice is not between a large hosted model and a small local one—it is between a small local model and no model at all. Serving cost then dominates, and the frozen BaiZe backbone [5] is built for exactly this regime.

Evaluation. We evaluate on held-out review items using expert agreement, and report the result beside the execution-based numbers of the text-only stage [5] so that verifiable and preference-based adaptation can be compared on the same system rather than in separate accounts.

Results are reported as **[TBD]**.

7 Conclusion

We built the multimodal core of BaiZe on the frozen BaiZe language backbone [5]: a vision encoder pre-trained from scratch, aligned to the backbone through a lightweight projector, and adapted to the semiconductor domain via preference-based reinforcement learning.

Through a controlled encoder comparison, we found that a larger ViT-L/14 tower (68.2% linear-probe, $+3.9$ pp over ViT-B/16) does *not* translate its advantage downstream: the aligned MLLM averages 26.1% over 11 benchmarks, indistinguishable from the smaller tower. We attribute the gap to the 224-px single-tile setting and the alignment architecture rather than to tower capacity.

Stages (ii) and (iii) are specified here as protocols with pre-specified criteria, evaluated across 12 general multimodal benchmarks, 4 domain benchmarks, and the multimodal subset of an EDA flow benchmark; their measurements are reported as **[TBD]**.

The immediate next step is closing the encoder-to-downstream transfer gap—through higher-resolution multi-tile input and domain-specific alignment—toward a deployable multimodal core for visual EDA tasks.

References

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